

1. THE INTERESTING OCTET

Definition: The first octet in the subnet mask from the left that is **NOT 255** and **NOT 0** [cite: 25, 29]. This is where active subnetting occurs [cite: 25].

Block Size (Increment) = 256 - Interesting Octet Value [cite: 26, 30]

Purpose: Determines subnet boundaries, Network Addresses, and Broadcast Addresses [cite: 27].

Example Breakdown

SUBNET MASK	INTERESTING OCTET	BLOCK SIZE
255.255.252.0	3rd Octet (252) [cite: 28]	256 - 252 = 4 [cite: 28]
255.255.255.192	4th Octet (192) [cite: 28]	256 - 192 = 64 [cite: 28]
255.224.0.0	2nd Octet (224) [cite: 28]	256 - 224 = 32 [cite: 28]

QUICK REFERENCE TABLE

PREFIX / MASK	INTERESTING OCTET	BLOCK SIZE
/24 (255.255.255.0)	None (4th) [cite: 31]	256 [cite: 31]
/25 (255.255.255.128)	128 (4th) [cite: 31]	128 [cite: 31]
/26 (255.255.255.192)	192 (4th) [cite: 31]	64 [cite: 31]
/27 (255.255.255.224)	224 (4th) [cite: 31]	32 [cite: 31]
/22 (255.255.252.0)	252 (3rd) [cite: 31]	4 [cite: 31]
/20 (255.255.240.0)	240 (3rd) [cite: 31]	16 [cite: 31]
/11 (255.224.0.0)	224 (2nd) [cite: 31]	32 [cite: 31]

2. SUBNET BOUNDARY RULES

Why 192.168.1.64/25 is Invalid:

A /25 network (mask 255.255.255.128) has a block size of **128** [cite: 32, 33]. Subnet boundaries must start at multiples of 128 [cite: 33, 34]:

- **Subnet 1:** 192.168.1.0/25 (Range: .0 – .127) [cite: 34]
- **Subnet 2:** 192.168.1.128/25 (Range: .128 – .255) [cite: 34]

✗ **192.168.1.64/25** falls inside Subnet 1; it is a host IP, not a valid Network ID [cite: 34].

Why /26 Starts at .64:

A /26 mask (255.255.255.192) yields block size **64** [cite: 35, 36]. Valid subnets start at increments of 64: **.0, .64, .128, .192** [cite: 36, 37].

Entire /24: 0-----255 [cite: 34]
/25 Split: 0-----127 | 128-----255 [cite: 34]
/26 Split: 0--63 | 64--127 | 128--191 | 192--255 [cite: 37]

3. SUBNETTING EXAMPLE (< /24)

Target: Solve for **172.16.10.45/23** [cite: 38]

1. **Subnet Mask:** /23 = **255.255.254.0** [cite: 38]
2. **Interesting Octet:** 3rd Octet (**254**) [cite: 38]
3. **Block Size:** 256 - 254 = **2** [cite: 38]
4. **Focus Value:** 3rd octet of IP is **10** [cite: 38]
5. **Network Multiplier:** Multiples of 2: 0, 2, 4, 6, 8, **10**, 12... [cite: 38] → Network ID starts at **172.16.10.0** [cite: 38]
6. **Next Network:** 10 + 2 = **172.16.12.0** [cite: 38]
7. **Broadcast Address:** One before next network = **172.16.11.255** [cite: 38]

Result Summary

PROPERTY	CALCULATED VALUE
Network Address	172.16.10.0 [cite: 38]
Broadcast Address	172.16.11.255 [cite: 38]
First Host IP	172.16.10.1 [cite: 38]
Last Host IP	172.16.11.254 [cite: 38]

4. UNIVERSAL SUBNETTING ALGORITHM

1. **Convert Prefix to Mask** (e.g., /26 → 255.255.255.192) [cite: 39].
2. **Identify Interesting Octet** (1st octet ≠ 255 and ≠ 0) [cite: 39].
3. **Calculate Block Size:** 256 - Interesting Octet [cite: 39].
4. **Find Network Address:** Highest multiple of Block Size ≤ IP's value in the interesting octet [cite: 39]. Reset right octets to 0 [cite: 38].
5. **Find Next Network:** Add Block Size to current Network ID [cite: 39].
6. **Find Broadcast:** Next Network - 1 [cite: 39].
7. **Host Range:** Network + 1 to Broadcast - 1 [cite: 39].

Prefix ≥ /24 → 4th Octet Subnetting [cite: 39]
Prefix < /24 → 3rd, 2nd, or 1st Octet Subnetting [cite: 39]

5. CORE RULES & FORMULA SUMMARY

- **Block Size Formula:** 256 - Interesting Octet [cite: 26, 39]
- **Next Network:** Current Network + Block Size [cite: 39]
- **Broadcast Address:** Next Network - 1 [cite: 39]
- **First Host:** Network Address + 1 [cite: 39]
- **Last Host:** Broadcast Address - 1 [cite: 39]